

TOPIC 5: FURTHER MECHANICS

5A: FURTHER MOMENTUM

5A.1 ENERGY IN COLLISIONS

- **Linear Momentum Conservation**

- **Core Concept:** Linear momentum (p) is always conserved in any collision between objects.
- **Governing Law:** This is a consequence of Newton's third law of motion.
- **Mechanism:** Newton's second law expresses that force is proportional to the rate of change of momentum ($F = \frac{\Delta p}{\Delta t}$).
- **Condition:** Total energy in all forms is always conserved, but kinetic energy specifically may not be.

- **Elastic Collisions**

- **Definition:** A collision in which total kinetic energy (E_k) is conserved.
- **Mechanism:** Velocity of the second ball after collision must be identical to the initial velocity of the first ball if masses are equal.
- **Special Case:** Perfectly elastic collisions are rare and usually caused by non-contact forces, such as alpha particles scattered by a nucleus. Newton's cradle is nearly perfectly elastic.

- **Inelastic Collisions**

- **Definition:** A collision in which total kinetic energy (E_k) is not conserved.
- **Mechanism:** Kinetic energy is transferred into other forms such as heat and sound.
- **Constraint:** The total momentum before and after the collision must remain identical regardless of E_k loss.

- **Particle Momentum and Kinetic Energy**

- **Governing Equation:** Combining $E_k = \frac{1}{2}mv^2$ and $p = mv$ results in:

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$$E_k = \frac{p^2}{2m}.$$

- **Domain of Validity:** Particularly useful for subatomic particles traveling at non-relativistic speeds (much slower than the speed of light).

- **Application:** Used at the Large Hadron Collider (CERN) to calculate the mass of unknown particles by colliding particles with known ones and detecting movements.

[Diagram: Figure E - Discovery of a muon particle from the collision of a neutron with an unknown stationary particle, showing momentum vectors before and after].

5A.2 MORE COLLISIONS

- **Two-Dimensional Collisions**

- **Core Concept:** Momentum is a vector quantity and is conserved in each dimension separately.
- **Mechanism:** Resolve momentum into right-angle vector components (x and y) before and after collision. Component vectors are recombined to give the overall resultant vector.
- **Equation (Component Resolution):**
 - $p_{parallel} = p \cos \theta$.
 - $p_{perpendicular} = p \sin \theta$.

- **Impulse**

- **Definition:** The product of a force (F) applied for a known time (Δt).
- **Governing Principle:** Impulse is equal to the change in momentum (Δp).
- **Equations:**
 - $\text{impulse} = F\Delta t$.
 - $F\Delta t = \Delta p = m\Delta v$.
- **Units:** Newton-seconds (Ns) or kilograms-metres per second ($kgms^{-1}$).

- **Applications: Asteroid Deflection (Thinking Bigger)**

- **Scenario:** Deflecting Near-Earth Objects (NEOs) using kinetic impactors.
- **Mechanism:** A spacecraft (impactor) crashes into an asteroid to change its momentum and trajectory.
- **Significant Results:** Small changes in trajectory (e.g., microdegrees) can result in position changes of thousands of kilometers over long distances (e.g., Earth's orbit).
- **Other Methods:** Slow gravity tractor, laser ablation, or nuclear detonation.

[Diagram: Figure B - Vector sum of total momentum in two dimensions showing p_{total} as the hypotenuse of a right-angled triangle formed by $p_{parallel}$ and $p_{perpendicular}$].

5B: CIRCULAR MOTION

5B.1 CIRCULAR MOTION BASICS

- **The Radian**
 - **Definition:** A unit of angle measurement where the angle (θ) is defined as the distance traveled (arc length, s) divided by the radius (r).
$$\theta = \frac{s}{r}.$$
 - **Equation:**
 - **Special Condition:** For a complete circle, $\theta = 2\pi$ radians (360°).
 - **Conversion:** 1 radian $\approx 57.3^\circ$.
 - **Angular Displacement (θ)**
 - **Definition:** The vector measurement of the angle through which an object has moved.
 - **Standard Convention:** Anticlockwise rotation is positive; clockwise rotation is negative.
 - **Angular Velocity (ω)**
 - **Definition:** The rate at which an object is moving around a circle.
 - **Governing Equation:** $\omega = \frac{\Delta\theta}{\Delta t}.$
 - **Units:** Radians per second (rads^{-1}).
 - **Periodic Motion:** For an object completing a full circle in time period (T):
 - $\omega = \frac{2\pi}{T}.$
 - $\omega = 2\pi f$ (where f is frequency).
 - **Instantaneous Velocity (v)**
 - **Definition:** The actual velocity of the moving object at any point on the circle, sometimes called tangential velocity.
 - **Mechanism:** Since $s = r\theta$, then $v = \frac{r\theta}{t}.$
 - **Governing Equation:** $v = r\omega.$
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5B.2 CENTRIPETAL FORCE

- **Centripetal Acceleration (a)**

- **Core Concept:** An object in a circle is constantly accelerating because its direction is constantly changing, even if speed is constant.
- **Direction:** Always directed towards the center of the circle.
- **Governing Equations:**
 - $a = \frac{v^2}{r}$.
 - $a = r\omega^2$.
- **Centripetal Force (F)**
 - **Definition:** The resultant force required to maintain an object's circular motion.
 - **Mechanism:** It is not a unique type of force; it is provided by other forces like tension, gravity, or friction.
 - **Governing Equations:**
 - $F = \frac{mv^2}{r}$.
 - $F = mr\omega^2$.
 - **Constraints:** Force is larger if mass (m) is greater, rotation is faster (ω or v), or radius (r) is smaller.
- **Application: Artificial Gravity (Thinking Bigger)**
 - **Definition:** The inertial reaction to the centripetal acceleration acting on a body in circular motion.
 - **Parameters:** Characterized by radius (r), angular velocity (ω), tangential velocity (v), and centripetal acceleration (a).
 - **Comfort Criteria:** High rotation speeds can cause motion sickness due to the Coriolis effect (distortion of apparent gravity during movement).
 - **Significance:** Necessary for long-term space missions to prevent weightlessness issues.

[Diagram: Figure C - Experimental detail for centripetal force experiment using a rubber bung, glass tube, and hanging masses to provide tension].

CONSOLIDATED EQUATION AND SYMBOL LIST

- **Momentum and Impulse**
 - $p = mv$ (Momentum)
 - $E_k = \frac{p^2}{2m}$ (Kinetic energy in terms of momentum)
 - impulse = $F\Delta t = \Delta p$ (Impulse-momentum theorem)
 - $p_x = p \cos \theta$; $p_y = p \sin \theta$ (Vector resolution)

- **Circular Motion**

- $\theta = \frac{s}{r}$ (Angle in radians)
- $\omega = \frac{\Delta\theta}{\Delta t}$ (Angular velocity)
- $\omega = \frac{2\pi}{T} = 2\pi f$ (Angular velocity for periodic motion)
- $v = r\omega$ (Tangential velocity)
- $a = \frac{v^2}{r} = r\omega^2$ (Centripetal acceleration)
- $F = \frac{mv^2}{r} = mr\omega^2$ (Centripetal force)

- **Symbols**

- p : Linear momentum ($kgms^{-1}$)
- E_k : Kinetic energy (J)
- F : Force (N)
- Δt : Time interval (s)
- θ : Angular displacement (rad)
- s : Arc length (m)
- r : Radius of circle (m)
- ω : Angular velocity ($rad s^{-1}$)
- v : Tangential/Instantaneous velocity (ms^{-1})
- a : Centripetal acceleration (ms^{-2})
- T : Time period (s)
- f : Frequency (Hz)

ULTRA-CONDENSED FINAL SUMMARY

- **Momentum:** Always conserved. In elastic collisions, E_k is also conserved; in inelastic collisions, E_k is lost to surroundings.
- **Impulse:** Equal to change in momentum ($F\Delta t = \Delta p$).
- **2D Collisions:** Resolve momentum into components, conserve separately, then recombine.
- **Radians:** $2\pi \text{ rad} = 360^\circ$. Angle is arc length over radius.
- **Angular Velocity (ω):** Rate of change of angle ($rad s^{-1}$); relates to tangential speed via $v = r\omega$.

- **Centripetal Motion:** A resultant force ($F = \frac{mv^2}{r}$) is always required to act toward the center of the circle to maintain circular motion.
- **Applications:** Use momentum conservation for particle physics and asteroid defense; use circular motion for centrifuges and artificial gravity.

TOPIC 6: ELECTRIC AND MAGNETIC FIELDS

6A ELECTRIC FIELDS

6A.1 ELECTRIC FIELDS

- **Definition: Electric Field**
 - A region of space where a charged particle experiences a force of electrical origin.
- **Governing Law: Electric Field Strength (E)**
 - Defined as the force-to-charge ratio for a charged particle experiencing a force due to an electric field.
 - Equation: $E = \frac{F}{Q}$.
 - Units: Newton per Coulomb (NC^{-1}) or Volts per metre (Vm^{-1}).
- **Conceptual Explanation: Field Lines and Equipotentials**
 - **Electric Field Lines:** Imaginary arrows showing the direction of force on a positive charge. Lines can never cross.
 - **Equipotentials:** Positions within a field with zero potential difference between them (constant potential). Always perpendicular to field lines.
- **Uniform Electric Fields**
 - Produced between parallel conducting plates at different electrical potentials.
 - Field is uniform if field lines are parallel across the whole field.
 - **Governing Equation:** $E = \frac{V}{d}$ where V is potential difference and d is distance between plates.
 - **Work and Energy:** Kinetic energy (E_k) gained by a charge moving through a potential difference (V): $E_k = VQ$.
- **Special Case: Millikan's Oil Drop Experiment**
 - **Objective:** To determine the elementary charge of an electron.
 - **Mechanism:** Balancing the weight of a charged oil droplet against the electric force in a uniform field.
 - **Governing Equations:**

- Weight: $W = mg$.
- Electric Force: $F = EQ = \frac{VQ}{d}$.
- At equilibrium: $mg = \frac{VQ}{d}$.
- **Stokes' Law (Viscous Drag):** Used to find the radius of the drop when falling at terminal velocity (v_{term}): $F = 6\pi\eta r v_{term}$.
- **Derived Result:** Charge Q is always a multiple of $1.602 \times 10^{-19}C$.

[Diagram: Figure C - Uniform electric field between parallel plates showing parallel field lines from + to - and dashed perpendicular equipotential lines] [Diagram: Figure G - Millikan's oil drop chamber showing atomiser, parallel plates with p.d. V , distance d , and viewing microscope]

6A.2 RADIAL ELECTRIC FIELDS

- **Definition: Radial Field**
 - The field around a point charge or a uniformly charged sphere where field lines act outwards (for positive) or inwards (for negative) in all directions from the centre.
- **Governing Law: Radial Field Strength**
 - Field strength reduces as distance from the centre increases.
 - **Equation:** $E = \frac{Q}{4\pi\epsilon_0 r^2}$.
 - **Constants:** ϵ_0 (permittivity of free space) = $8.85 \times 10^{-12} Fm^{-1}$; Coulomb's constant $k = \frac{1}{4\pi\epsilon_0} \approx 8.99 \times 10^9 Nm^2C^{-2}$.
- **Radial Electric Potential (V)**
 - The energy per unit charge at a distance r from charge Q .
 - **Equation:** $V = \frac{Q}{4\pi\epsilon_0 r}$.
 - **Constraint:** Zero potential is defined at infinity ($r = \infty$).
- **Potential Gradient**
 - Field strength E equals the rate of change of potential V with distance x .
 - **Equation:** $E = -\frac{dV}{dx}$.
- **Special Cases: Combined Fields**
 - Overall field is the vector sum of contributions from each individual charge.
 - Charge concentrates at spikes or points, creating very high field strengths and potential for charge leakage (lightning conductors).

[Diagram: Figure A - Radial electric field around a positive point charge with field lines diverging outwards and concentric equipotential circles]

6A.3 COULOMB'S LAW

- **Governing Law**
 - The force (F) between two point charges (Q_1, Q_2) is proportional to the product of the charges and inversely proportional to the square of the distance (r) between them.
- **Equation:**
$$F = \frac{Q_1 Q_2}{4\pi\epsilon_0 r^2} = \frac{kQ_1 Q_2}{r^2}.$$
- **Assumptions and Limitations**
 - Valid for point charges or spheres (distance measured between centres).
 - In a vacuum (air is treated as near-vacuum).
- **Mechanistic Reasoning: Attraction and Repulsion**
 - Force is a vector. A negative value of F (opposite charge signs) indicates attraction; a positive value indicates repulsion.
- **Practical Significance: Everyday Levitation**
 - Normal contact force is actually the macroscopic manifestation of electrostatic repulsion between surface atoms.

[Diagram: Figure D - Forces on a hanging charged ball resolved into components: Tension $T \sin \theta = F_{coulomb}$ and $T \cos \theta = mg$]

6A THINKING BIGGER: POWER LINE PROBLEMS

- **Core Concept:** Transmission lines carry very high voltages (e.g., $500kV$), generating significant electric fields.
 - **Safety Mechanism: Shielding**
 - If a conductor is inside a grounded conducting enclosure, the field outside is zero.
 - **Biological Impact:** Electric fields exert forces on charges in objects/tissue. At a line's surface, field strength is high (approx. $27kV/m$); inside the human body, it is much lower (approx. $0.008V/m$ in the torso) due to conductivity.
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6B CAPACITORS

6B.1 CAPACITOR BASICS

- **Definition: Capacitor**
 - An electrical circuit component that stores charge and electrical potential energy.
- **Governing Law: Capacitance (C)**
 - A measure of the capability of a capacitor; the amount of charge stored per unit voltage.
 - **Equation:** $C = \frac{Q}{V}$.
 - **Units:** Farads (F); usually measured in $\mu F, nF, pF$.
- **Energy Stored on a Capacitor (W)**
 - Calculated as the area under a Potential Difference (V) vs. Charge (Q) graph.
 - **Governing Equations:**
 - $W = \frac{1}{2}QV$.
 - $W = \frac{1}{2}CV^2$.
 - $W = \frac{Q^2}{2C}$.
- **Mechanism: Storing Charge**
 - Electrons flow from the negative terminal to one plate, making it negative. This repels electrons from the opposite plate back to the positive terminal, leaving it positive. The insulator (dielectric) prevents electrons from crossing the gap.

[Graph: Figure D - Potential difference (V) vs. Charge (Q) showing a straight line through the

origin; shaded area represents stored energy $W = \frac{1}{2}QV$]

6B.2 CHARGING AND DISCHARGING CAPACITORS

- **Mechanism: Discharge**
 - Current is highest at the start ($I = \frac{V}{R}$) and decreases as charge (and thus potential difference) on the capacitor falls.
- **The Time Constant (τ)**
 - The product of capacitance (C) and resistance (R). Represents the time taken for the current, p.d., or charge to fall to 37% ($1/e$) of its initial value.
 - **Equation:** $\tau = RC$.
 - **Significance:** Higher R or C increases the time taken to charge/discharge.
- **Special Case: Charging**

- P.D. and Charge increase toward supply maximum; current decreases exponentially toward zero.

[Graph: Figure B - Exponential decay curves for (a) Discharge current, (b) p.d. across capacitor, and (c) Charge on capacitor over time]

6B.3 CAPACITOR MATHEMATICS

- **Governing Equations: Exponential Decay**
 - **Charge:** $Q = Q_0 e^{-t/RC}$.
 - **Voltage:** $V = V_0 e^{-t/RC}$.
 - **Current:** $I = I_0 e^{-t/RC}$.
 - **Logarithmic Linearization**
 - Used to determine RC from experimental data.
 - **Equations:**
 - $\ln Q = \ln Q_0 - \frac{t}{RC}$.
 - $\ln V = \ln V_0 - \frac{t}{RC}$.
 - $\ln I = \ln I_0 - \frac{t}{RC}$.
 - **Graphing:** A plot of $\ln V$ against t yields a straight line with gradient $-\frac{1}{RC}$.
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6B THINKING BIGGER: ULTRACAPACITORS

- **Definition:** High-capacity capacitors (e.g., $100F g^{-1}$) often using carbon nanotubes or graphene composites.
 - **Significance:** Potential replacement for electrochemical batteries in electric cars due to rapid charging/discharging capabilities.
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6C ELECTROMAGNETIC EFFECTS

6C.1 MAGNETIC FIELDS

- **Definitions**
 - **Magnetic Field:** A region of space that will cause a magnetic pole to feel a force.

- **Magnetic Flux (Φ):** The quantity of "magnetism" passing through an area. Unit: Weber (Wb).
 - **Magnetic Flux Density (B):** The strength of the magnetic field; flux per unit area. Unit: Tesla (T).
 - **Governing Equations**
 - Flux: $\Phi = BA \sin \theta$ (where θ is the angle between the field and the plane of the area).
 - Flux Linkage: Total flux through a coil of N turns. $N\Phi = BAN \sin \theta$.
 - **Mechanism**
 - Field lines show direction a North pole would be pushed (North to South). Closer lines indicate higher flux density (B).
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6C.2 ELECTRIC MOTORS

- **The Motor Effect**
 - A wire carrying a current, held within a magnetic field, will experience a force.
 - **Governing Rule: Fleming's Left Hand Rule**
 - **Thumb:** Motion/Force (F).
 - **First Finger:** Field (B , North to South).
 - **Second Finger:** Current (I , Positive to Negative).
 - **Condition:** If current is parallel to the magnetic field, there is zero force.
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6C.3 MAGNETIC FORCES

- **Forces on Wires**
 - **Equation:** $F = BIL \sin \theta$.
 - Maximum force when $\theta = 90^\circ$ ($\sin \theta = 1$).
- **Forces on Charged Particles**
 - **Equation:** $F = Bqv \sin \theta$.
 - **Mechanism:** Since the force is always at right angles to velocity, it acts as a **centripetal force**, causing particles to move in circular paths.
- **Special Case: The Mass Spectrometer**
 - Used to identify chemicals by their charge-to-mass ratio (q/m).
 - **Derivation:**
 - Centripetal force: $F = \frac{mv^2}{r}$.
 - Magnetic force: $F = Bqv$.
 - Equating: $Bqv = \frac{mv^2}{r} \Rightarrow \frac{q}{m} = \frac{v}{Br}$.

- Velocity v is often provided by an electric field: $\frac{1}{2}mv^2 = qV \Rightarrow v = \sqrt{\frac{2qV}{m}}$.
- Final Form:** $\frac{q}{m} = \frac{2V}{B^2r^2}$.

[Diagram: Figure B - Charged particles in a magnetic field following a circular path as force is perpendicular to motion] [Diagram: Figure D - Schematic of a mass spectrometer showing ionisation, acceleration, deflection by electromagnet, and detection]

6C.4 GENERATING ELECTRICITY

- Electromagnetic Induction**
 - Moving a conductor in a magnetic field, or changing the field around a conductor, induces an electromotive force (e.m.f.).
 - Governing Law: Faraday's Law**
 - Induced e.m.f. (ϵ) is proportional to the rate of change of magnetic flux linkage ($N\Phi$).
 - Equation:** $\epsilon = -\frac{d(N\Phi)}{dt} = -\frac{\Delta(N\Phi)}{\Delta t}$.
 - Governing Law: Lenz's Law**
 - The direction of an induced e.m.f. is such as to oppose the change creating it. Represented by the negative sign in Faraday's equation.
 - Practical Application: Transformers**
 - Changing current in the primary coil creates a changing magnetic field in the core, inducing a changing e.m.f. in the secondary coil.
 - Equation:** $\frac{V_p}{V_s} = \frac{N_p}{N_s}$.
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CONSOLIDATED EQUATION AND SYMBOL LIST

- Electric Fields**
 - $E = \frac{F}{Q}$ (Field strength definition)
 - $E = \frac{V}{d}$ (Uniform field)
 - $E_k = VQ$ (Work done on a charge)

- $E = \frac{Q}{4\pi\epsilon_0 r^2}$ (Radial field strength)
- $V = \frac{Q}{4\pi\epsilon_0 r}$ (Radial potential)
- $E = -\frac{dV}{dx}$ (Potential gradient)
- $F = \frac{Q_1 Q_2}{4\pi\epsilon_0 r^2}$ (Coulomb's Law)
- **Capacitors**
 - $C = \frac{Q}{V}$ (Capacitance definition)
 - $W = \frac{1}{2} QV = \frac{1}{2} CV^2 = \frac{Q^2}{2C}$ (Stored energy)
 - $\tau = RC$ (Time constant)
 - $Q = Q_0 e^{-t/RC}$ (Discharge charge)
 - $V = V_0 e^{-t/RC}$ (Discharge voltage)
 - $I = I_0 e^{-t/RC}$ (Discharge current)
 - $\ln Q = \ln Q_0 - \frac{t}{RC}$ (Logarithmic discharge)
- **Magnetic Fields**
 - $\Phi = BA \sin \theta$ (Magnetic flux)
 - $B = \frac{\Phi}{A}$ (Flux density)
 - Flux linkage $= N\Phi = BAN \sin \theta$
 - $F = BIL \sin \theta$ (Force on wire)
 - $F = Bqv \sin \theta$ (Force on particle)
 - $r = \frac{mv}{Bq}$ (Radius of circular path)
 - $\frac{q}{m} = \frac{2V}{B^2 r^2}$ (Specific charge in mass spec)
 - $\epsilon = -\frac{d(N\Phi)}{dt}$ (Faraday's Law)
 - $\frac{V_p}{V_s} = \frac{N_p}{N_s}$ (Transformer ratio)

ULTRA-CONDENSED FINAL SUMMARY

- **Electric Fields:** Strength E is force per unit charge. Uniform (V/d) vs Radial ($1/r^2$). Potential (V) is energy per unit charge ($1/r$).
- **Capacitors:** Store charge ($Q = CV$) and energy ($1/2 CV^2$). Discharge is exponential; time constant $\tau = RC$ governs rate.
- **Magnetic Fields:** Flux density B is strength (T). Flux linkage $N\Phi$ induces e.m.f. per Faraday's Law ($\epsilon = -\Delta N\Phi/\Delta t$).
- **Forces:** Magnetic forces (BIL , Bqv) are perpendicular to field and current/velocity. Particles move in circles ($r = mv/Bq$).
- **Induction:** Lenz's Law ensures induction opposes change. Transformers change voltage via turn ratios.

TOPIC 7: NUCLEAR AND PARTICLE PHYSICS

7A PROBING MATTER

7A.1 A NUCLEAR ATOM

- **Atomic Theories Through History**
 - **Democritus (~400 BCE):** Suggested existence of atoms as smallest indivisible units.
 - **Dalton (1803):** Atoms have different masses and combine to produce compound molecules.
 - **Thomson (1906) "Plum Pudding Model":** Atom composed of a positively charged sphere ("dough") with tiny negatively charged electrons ("plums") scattered through it.
 - **Rutherford (1911):** Nuclear model; most mass concentrated in a tiny central nucleus, with electrons orbiting at a distance through empty space.
 - **Bohr (1913):** Miniature Solar System; electrons limited to fixed orbits/quantised energy levels.
 - **Quantum Mechanical Model (1920s):** Heisenberg's uncertainty principle implies exact positions/velocities cannot be known; electrons exist in "probability clouds" or "orbitals".
- **Alpha Particle Scattering (Geiger-Marsden Experiment)**
 - **Core Concept:** Using alpha particles (α) as probes to investigate the structure of gold atoms in a thin foil (10^{-6} m).
 - **Governing Law:** Conservation of momentum and electrostatic repulsion.

- **Mechanism:** Alpha particles fired at gold foil in a vacuum. Scintillations detected on a zinc sulfide screen via microscope.
- **Observations and Conclusions:**
 1. **Most (99.9%) pass straight through:** Most of the atom is empty space.
 2. **Small fraction deflected at small angles (10° to 90°):** Large concentration of charge in one place.
 3. **Very few (1 in 8000) deflected back ($> 90^\circ$):** Most mass and charge concentrated in a tiny, central nucleus.
- **Condition:** Vacuum is necessary so alpha particles do not collide with air atoms.
- **Chadwick's Discovery of the Neutron (1932)**
 - **Conceptual Explanation:** Rutherford noted a discrepancy between nuclear mass and proton count.
 - **Mechanism:** Alpha particles hitting beryllium emitted radiation that removed protons from paraffin (detected by Geiger-Müller tube).
 - **Significance:** Radiation was a neutral particle (n) with mass approx. 1% greater than a proton.
- **Nuclear Structure**
 - **Nucleons:** Collective term for protons (p) and neutrons (n).
 - **Proton Number (Z):** Atomic number; defines the element.
 - **Nucleon Number (A):** Mass number; total protons + neutrons ($A = Z + N$).
 - **Isotopes:** Atoms of the same element with different neutron numbers.
 - **Strong Nuclear Force:** Extremely short-range force between hadrons that holds the nucleus together against electrostatic repulsion of protons.
 - **Stability:** Above $Z = 20$, more neutrons are needed for stability; big nuclei may have $> 50\%$ more neutrons than protons.

[Diagram: Figure B - Rutherford's scattering apparatus showing alpha source, gold foil, and rotating microscope with ZnS screen in a vacuum] [Diagram: Figure C - Alpha particle paths near a gold nucleus: straight through, small deflection, and large-angle back-scattering]

7A.2 ELECTRONS FROM ATOMS

- **Thermionic Emission**
 - **Definition:** The release of electrons from a metal surface due to heating.
 - **Mechanism:** Free conduction electrons gain enough thermal energy to escape the metal's surface.
 - **Significance:** Principle used in cathode ray tubes (CRTs) to produce electron beams.

- **Electron Acceleration**

- **Core Concept:** Electrons in an electric field gain kinetic energy (E_k).

- **Governing Equation:** $E_k = eV = \frac{1}{2}mv^2$.

- **Derivation for Velocity (v):**

- $v = \sqrt{\frac{2eV}{m}}$.

- **Constants:** $e \approx 1.6 \times 10^{-19}$ C; $m_e \approx 9.11 \times 10^{-31}$ kg.

- **An Electron Probe**

- **Definition: de Broglie Wavelength (λ):** The wavelength associated with a particle, inversely proportional to its momentum (p).

- **Governing Equation:** $\lambda = \frac{h}{p} = \frac{h}{mv}$.

- **Mechanism:** Electron beams fired at crystals produce diffraction patterns. Diffraction occurs if the wavelength is comparable to the spacing between atoms (10^{-10} m).

- **Condition:** Higher energies (E_k) produce smaller wavelengths (λ), allowing the probing of smaller structures like nucleons.

[Diagram: Figure B - Practical setup measuring electron diffraction through a thin piece of carbon to confirm de Broglie's hypothesis]

7B PARTICLE ACCELERATORS AND DETECTORS

7B.1 PARTICLE ACCELERATORS

- **Linear Accelerators (LINAC)**

- **Mechanism:** Accelerates particles (e.g., electrons) in a straight line through a series of hollow cylindrical electrodes.
- **Governing Law:** Electrostatic attraction/repulsion via high-frequency alternating voltage.
- **Constraints:**
 - Cylinders get progressively longer so particles spend the same time in each (as velocity increases).
 - Frequency of alternating voltage (f) must be constant (often gigahertz).

- **Special Case: Bertozzi's Experiment (1964):** Demonstrated that as $v \rightarrow c$, kinetic energy increases without limit but velocity does not; mass appears to increase (relativistic effects).
- **The Cyclotron**
 - **Definition:** A circular accelerator using two D-shaped electrodes ("dees") in a magnetic field.
 - **Mechanism:**
 - Magnetic field (B) causes circular motion ($F = Bqv$ acting as centripetal force).
 - Electric field in the gap between dees accelerates the particle twice per orbit.
 - **Derivation of Radius (r):**
 - $$Bqv = \frac{mv^2}{r} \Rightarrow r = \frac{mv}{Bq} = \frac{p}{Bq}.$$
 - **Cyclotron Frequency (f) and Time Period (T):**
 - $$T = \frac{2\pi r}{v} = \frac{2\pi m}{Bq}.$$
 - $$f = \frac{1}{T} = \frac{Bq}{2\pi m}.$$
 - **Significance:** Frequency is independent of radius; a constant frequency alternating p.d. can be used.
 - **Limitation:** Fails when particles reach relativistic speeds as mass (m) increases, changing the required frequency.
- **The Synchrotron**
 - **Definition:** A ring accelerator where both the magnetic field and alternating frequency are varied to keep particles in a fixed-radius path.
 - **Significance:** Can reach much higher energies (e.g., 1 TeV).

[Diagram: Figure A - Structure of a linear accelerator showing alternating polarity of cylindrical electrodes] [Diagram: Figure E - The structure of a cyclotron showing dees, magnetic field direction, and the spiraling outward path of a particle]

7B.2 PARTICLE DETECTORS

- **Principles of Detection: Ionisation**
 - **Mechanism:** Particles passing through a gas ionise atoms. The resulting ions and electrons are accelerated by an electric field toward electrodes, creating a measurable pulse of electricity.
 - **Geiger-Müller (GM) Detector:** Uses ionisation in argon gas to count particles.

- **Bubble Chambers**

- **Mechanism:** Uses superheated liquid (typically hydrogen). Ions produced by particles act as nucleation points for bubbles, leaving visible tracks.
- **Interpretation:**
 - **Curvature:** Path radius ($r = \frac{mv}{Bq}$) identifies mass and charge.
 - **Spiral:** Particles lose kinetic energy through ionisation, causing the radius to decrease.
 - **V-shaped tracks:** Indicate neutral particles decaying into charged pairs (neutral particles leave no track).

[Diagram: Figure A - Geiger-Müller detector showing mica window, argon gas, central anode, and outer cathode] [Diagram: Figure E - Creation of an electron-positron pair in a magnetic field showing opposite spiral directions]

7B.3 THE LARGE HADRON COLLIDER (LHC)

- **Core Details:** Giant synchrotron over 8 km in diameter, 100 m underground.
 - **Operating Parameters:**
 - Protons collide at energies of 14 TeV (14×10^{12} eV).
 - Velocity: 99.9999991% the speed of light.
 - Temperature: 1.9 K (colder than deep space).
 - **Major Experiments:**
 - **CMS (Compact Muon Solenoid) & ATLAS:** General-purpose; discovered Higgs boson in 2012.
 - **LHCb:** Investigates bottom quarks and matter/anti-matter asymmetry.
 - **ALICE:** Collides lead ions to study quark-gluon plasma.
-

7C THE PARTICLE ZOO

7C.1 PARTICLE INTERACTIONS

- **Creation and Annihilation**
 - **Governing Law:** Mass-energy equivalence ($E = mc^2$).
 - **Pair Production:** A high-energy photon (γ) converts into a particle and its anti-particle.
 - **Annihilation:** A particle and its anti-particle meet and convert all mass into energy (photons).
- **Units**

- **Electronvolt (eV):** Energy gained by an electron through 1 V (1.60×10^{-19} J).
 - **Mass units:** MeV/c^2 or GeV/c^2 .
 - **Atomic mass unit (u):** 1.66×10^{-27} kg $\approx$ 931.5 MeV/ c^2 .
-

7C.2 THE PARTICLE ZOO

- **The Standard Model**
 - **Quarks:** 6 types (flavours) in 3 generations.
 1. u (up), d (down): Charge $+\frac{2}{3}e, -\frac{1}{3}e$.
 2. c (charm), s (strange): Charge $+\frac{2}{3}e, -\frac{1}{3}e$.
 3. t (top), b (bottom): Charge $+\frac{2}{3}e, -\frac{1}{3}e$.
 - **Leptons:** Do not experience strong nuclear force.
 1. e^- (electron), ν_e (electron neutrino).
 2. μ (muon), ν_μ (muon neutrino).
 3. τ (tau), ν_τ (tau neutrino).
 - **Anti-particles:** Same mass, opposite charge/quantum numbers (e.g., positron e^+).
-

7C.3 PARTICLES AND FORCES

- **Hadrons:** Experience strong nuclear force.
 - **Baryons:** Composed of 3 quarks (e.g., Proton: uud ; Neutron: udd).
 - **Mesons:** Composed of a quark-antiquark pair (e.g., Pion π^+ : $u\bar{d}$; Kaon K^0 : $d\bar{s}$).
 - **Exchange Bosons (Force Carriers):**
 - **Electromagnetic:** Photon (γ).
 - **Strong:** Gluon (g).
 - **Weak:** W^+ , W^- , Z^0 bosons.
 - **Gravity:** Graviton (theoretical).
-

7C.4 PARTICLE REACTIONS

- **Conservation Laws:**
 1. **Charge (Q):** Always conserved.
 2. **Baryon Number (B):** Quarks have $B = +1/3$; Baryons have $B = +1$; Mesons/Leptons have $B = 0$.

3. **Lepton Number (L):** Conserved by generation (L_e, L_μ, L_τ). Electrons/Neutrinos have $L = +1$.
 4. **Strangeness (S):** Strange quark (s) has $S = -1$. Conserved in Strong/EM interactions; can change by ± 1 in Weak interactions.
- **Relativistic Lifetimes:** Particles moving near c have longer lifetimes as observed from Earth due to time dilation (e.g., cosmic ray muons).
-

CONSOLIDATED EQUATION AND SYMBOL LIST

- **Probing Matter**

- $E_k = eV = \frac{1}{2}mv^2$ (Electron acceleration)
- $\lambda = \frac{h}{p} = \frac{h}{mv}$ (de Broglie wavelength)
- $A = Z + N$ (Nucleon number)

- **Accelerators**

- $r = \frac{mv}{Bq}$ (Radius of circular path)
- $f = \frac{Bq}{2\pi m}$ (Cyclotron frequency)
- $T = \frac{2\pi m}{Bq}$ (Cyclotron time period)

- **Particle Physics**

- $\Delta E = c^2 \Delta m$ (Mass-energy equivalence)
- $E = hf$ (Photon energy)
- $1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$
- $1 \text{ u} = 931.5 \text{ MeV}/c^2$

- **Symbols**

- V : Potential difference (V)
- e : Elementary charge ($1.6 \times 10^{-19} \text{ C}$)
- h : Planck constant ($6.63 \times 10^{-34} \text{ J s}$)
- B : Magnetic flux density (T)
- m : Mass (kg)
- f : Frequency (Hz)
- c : Speed of light ($3.00 \times 10^8 \text{ m s}^{-1}$)

ULTRA-CONDENSED FINAL SUMMARY

- **Atomic Structure:** Nucleus (protons/neutrons) is tiny (10^{-15} m), dense, and positive. Confirmed by Rutherford's scattering.
- **Strong Force:** Holds nucleus together; overcomes EM repulsion. Short range (< 3 fm).
- **Duality:** Particles have wave properties; $\lambda = h/mv$. Small λ probes small structures.
- **Accelerators:** LINACs (linear), Cyclotrons (circular, constant f), Synchrotrons (variable B and f).
- **Standard Model:** Hadrons (Baryons = 3 quarks; Mesons = $q\bar{q}$) and Leptons. Forces carried by exchange bosons.
- **Conservations:** Charge, Baryon number, and Lepton number are absolute. Strangeness is only conserved in Strong interactions.

TOPIC 8: THERMODYNAMICS

8A HEAT AND TEMPERATURE

8A.1 HEAT AND TEMPERATURE

- **Kinetic Theory**
 - **Core Concept:** The idea that consideration of the microscopic movements of particles will predict the macroscopic behaviour of a substance.
- **Temperature Scales**
 - **Scales:** Celsius ($^{\circ}\text{C}$), Fahrenheit ($^{\circ}\text{F}$), Kelvin (K), and Rankine (R).
 - **Celsius Scale:** A scale with zero degrees at the freezing point of water and 100 degrees at the boiling point of water.
 - **Kelvin Scale:** An absolute temperature scale with each degree the same size as those on the Celsius scale.
 - **Absolute Temperature:** A temperature scale that starts at absolute zero.
- **Absolute Zero**
 - **Definition:** The theoretical temperature at which molecules will no longer be moving; all kinetic energy has been removed.
 - **Values:** 0 K or -273.15°C .
- **Temperature Conversions**
 - **Celsius to Kelvin:** $T \text{ (in K)} = \theta \text{ (in } ^{\circ}\text{C)} + 273.15$.
 - **Kelvin to Celsius:** $\theta \text{ (in } ^{\circ}\text{C)} = T \text{ (in K)} - 273.15$.
- **Practical Application: Thermostats**

- **Mechanism:** Potential divider circuits use a sensor component (like a thermistor) to control a second circuit (e.g., heating or cooling).
- **Behavior:** As temperature rises, thermistor resistance (R) changes, altering output voltage (V_{out}) to trigger motors or fans.

[Diagram: Figure C - Circuit diagram of a thermostat using a potential divider with a variable resistor, thermistor, and motor]

8A.2 HEAT TRANSFER

- **Specific Heat Capacity (c)**
 - **Definition:** The energy required to raise the temperature of 1 kg of a substance by one degree Kelvin (1 K).
 - **Governing Equation:** $\Delta E = mc\Delta\theta$.
 - **Units:** $J\ kg^{-1}\ K^{-1}$ or $J\ kg^{-1}\ ^\circ C^{-1}$.
 - **Mechanism:** Transferring heat increases the internal kinetic energy of molecules, causing them to move faster.
 - **Variables:** Temperature rise depends on mass (m), energy transferred (ΔE), and the material's specific heat capacity (c).
- **Specific Latent Heat (L)**
 - **Definition:** The energy required to change the state of 1 kg of a substance at a constant temperature.
 - **Governing Equation:** $\Delta E = L\Delta m$.
 - **Units:** $J\ kg^{-1}$.
 - **Types:**
 - **Specific Latent Heat of Fusion (L_f):** Energy for melting/freezing.
 - **Specific Latent Heat of Vaporisation (L_v):** Energy for boiling/condensing.
 - **Mechanism:** Energy is used to break bonds/increase molecular potential energy rather than increasing kinetic energy; temperature remains constant during phase change.
- **Potential Energy in Phase Change**
 - **Concept:** Molecules gain potential energy as they move apart.
 - **Sign Convention:** Potential energy is negative; as molecules move apart, potential energy increases (gets closer to zero).

[Diagram: Figure A - Experimental setup for measuring specific heat capacity with an immersion heater, thermometer, and insulated block] [Diagram: Figure B - Experimental setup for measuring specific latent heat of fusion using a funnel of crushed ice and a joulemeter]

8A.3 INTERNAL ENERGY

- **Internal Energy**
 - **Definition:** The sum of the kinetic and potential energies of all the molecules within a given mass of a substance.
 - **Distribution:** Energy is randomly distributed across all molecules.
- **Maxwell-Boltzmann Distribution**
 - **Definition:** A mathematical function that describes the distribution of energies (or speeds) amongst particles at a given temperature.
 - **Characteristics:**
 - Zero molecules have zero energy.
 - Only a few molecules have high energy.
 - There is no maximum value for the energy a molecule can have.
 - As temperature increases, the peak moves toward higher energies/speeds.
- **Root-Mean-Square Speed (c_{rms})**
 - **Definition:** The square root of the arithmetic mean value of the squares of the speeds of particles in an ideal gas.
 - **Symbol:** $\sqrt{\langle c^2 \rangle}$.
 - **Calculation:** Square individual speeds, find their mean, and then take the square root.
- **Molecular Kinetic Energy**
 - **Governing Theory:** Kinetic theory states average kinetic energy is proportional to absolute temperature (T).
 - **Governing Equation:** $\frac{1}{2}m \langle c^2 \rangle = \frac{3}{2}kT$.
 - **Boltzmann Constant (k):** $1.38 \times 10^{-23} \text{ J K}^{-1}$.
 - **Limitation:** At absolute zero (0 K), rms speed must decrease to zero.

[Graph: Figure A - Maxwell-Boltzmann distribution showing Number of Molecules vs. Energy/Speed at a specific temperature]

8A.4 IDEAL GAS BEHAVIOUR

- **Boyle's Law**
 - **Definition:** For a constant mass of gas at a constant temperature, the pressure (p) exerted by the gas is inversely proportional to the volume (V) it occupies.
 - **Proportionality:** $p \propto \frac{1}{V}$ or $pV = \text{constant}$.
- **Charles's Law**
 - **Definition:** For a constant mass of gas at a constant pressure, the volume (V) occupied by the gas is proportional to its absolute temperature (T).

- **Proportionality:** $V \propto T$.
- **The Pressure Law**
 - **Definition:** For a constant mass of gas at a constant volume, the pressure (p) exerted by the gas is proportional to its absolute temperature (T).
 - **Proportionality:** $p \propto T$.
- **Ideal Gases**
 - **Definition:** A theoretical gas which does not suffer from the real-world difficulties that mean real gases do not perfectly follow the gas laws.
 - **Assumptions (6):**
 1. Molecules have negligible size/volume.
 2. Molecules are identical.
 3. All collisions are perfectly elastic; time of collision is negligible.
 4. Molecules exert no forces on each other (except during collision).
 5. Motion is random.
 6. Enough molecules exist for statistics to be applied.
- **Equation of State**
 - **Definition:** The single equation that defines a gas in terms of its pressure, volume, temperature, and quantity.
 - **Form for Molecules:** $pV = NkT$ (where N is number of molecules).
 - **Form for Moles:** $pV = nRT$ (where n is number of moles).
- **The Mole (n)**
 - **Definition:** The SI unit for amount of substance; contains 6.02×10^{23} molecules (N_A).
 - **Universal Gas Constant (R):** $8.31 \text{ J kg}^{-1} \text{ mol}^{-1}$.
 - **Relation:** $R = N_A k$.
- **Average Molecular Kinetic Energy (Derivation Result)**
 - **Equation:** $pV = \frac{1}{3}Nm \langle c^2 \rangle$.
 - **Result:** $\frac{1}{2}m \langle c^2 \rangle = \frac{3}{2}kT$.

[Graph: Figure B - Pressure vs. 1/Volume showing a straight line through origin for Boyle's Law]

[Graph: Figure C - Volume vs. Absolute temperature showing a straight line through origin for Charles's Law]

[Graph: Figure E - Pressure vs. Absolute temperature showing a straight line through origin for Pressure Law]

8A THINKING BIGGER: REACH FOR THE SKY!

- **Concept:** Gas-filled balloons (helium or hydrogen) use buoyancy (less dense than air) to rise.
- **Physics:** As altitude increases, atmospheric pressure decreases, causing the balloon to expand (Boyle's Law).

- **History:** Montgolfier brothers used hot air (phlogiston theory, now obsolete); modern balloons use density differences.
-

CONSOLIDATED EQUATION AND SYMBOL LIST

- **Temperature**
 - $T \text{ (K)} = \theta \text{ (}^\circ\text{C)} + 273.15$
 - **Heat Transfer**
 - $\Delta E = mc\Delta\theta$ (Specific Heat Capacity)
 - $\Delta E = L\Delta m$ (Specific Latent Heat)
 - **Kinetic Theory and Internal Energy**
 - $E_{int} = \sum E_k + \sum E_p$ (Internal Energy)
 - $\frac{1}{2}m \langle c^2 \rangle = \frac{3}{2}kT$ (Molecular Kinetic Energy)
 - $c_{rms} = \sqrt{\langle c^2 \rangle}$ (rms speed)
 - **Ideal Gases**
 - $pV = NkT$ (Ideal gas law, molecules)
 - $pV = nRT$ (Ideal gas law, moles)
 - $pV = \frac{1}{3}Nm \langle c^2 \rangle$ (Pressure from kinetic theory)
 - $n = \frac{N}{N_A}$ (Mole calculation)
 - **Symbols**
 - c : Specific heat capacity ($\text{J kg}^{-1} \text{ K}^{-1}$)
 - L : Specific latent heat (J kg^{-1})
 - k : Boltzmann constant ($1.38 \times 10^{-23} \text{ J K}^{-1}$)
 - R : Universal gas constant ($8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)
 - N_A : Avogadro's number (6.02×10^{23})
 - T : Absolute temperature (K)
 - p : Pressure (Pa)
 - V : Volume (m^3)
-

ULTRA-CONDENSED FINAL SUMMARY

- **Absolute Zero:** 0 K (-273.15°C), where all molecular kinetic energy is zero.
- **Internal Energy:** Sum of random molecular kinetic and potential energy. Phase changes involve potential energy change at constant T .

- **Gas Laws:** Boyle ($pV = \text{const}$), Charles ($V/T = \text{const}$), Pressure ($p/T = \text{const}$).
- **Ideal Gas Equation:** $pV = nRT$. Assumptions include negligible particle volume and no intermolecular forces.
- **Molecular Energy:** Average $E_k = \frac{3}{2}kT$. Peak of Maxwell-Boltzmann distribution shifts right with higher T .

TOPIC 9: NUCLEAR DECAY

9A RADIOACTIVITY

9A.1 NUCLEAR RADIATION

- **Background Radiation**
 - **Definition:** Low levels of nuclear radiation that human beings are continually exposed to from natural and environmental sources.
 - **Governing Unit:** The becquerel (Bq), where $1 \text{ Bq} = 1 \text{ count per second}$.
 - **Sources of Background Radiation:**
 - Radon gas (50.0%).
 - Ground and buildings (14.0%).
 - Medical (14.0%).
 - Food and drink (11.5%).
 - Cosmic rays (10.0%).
 - Nuclear power and weapons (0.3%).
 - Other (0.2%).
 - **Practical Significance:** In experiments, background radiation must be measured separately and subtracted from the main count to obtain the "corrected count".
- **Types of Nuclear Radiation**
 - **Core Concept:** Nuclei decay to change from one type to another, or change their composition/energy state, emitting radiation carrying kinetic energy.
 - **Alpha Decay (α):**
 - **Definition:** Emission of a helium nucleus consisting of two protons and two neutrons.
 - **Properties:** Positive charge ($+2e$); relatively large mass; highly ionising (10,000 ionisations per mm).
 - **Penetration:** Low; range of few cm in air; blocked by paper or skin.
 - **Governing Equation:** ${}^{241}_{95}\text{Am} \rightarrow {}^{237}_{93}\text{Np} + {}^4_2\alpha$.
 - **Beta Decay (β):**

- **Definition:** Emission of an electron (e^-) from the nucleus when a neutron decays into a proton.
 - **Properties:** Single negative charge ($-1e$); much smaller size; less ionising than alpha.
 - **Penetration:** Moderate; range of few cm to several meters in air; blocked by thin aluminium sheet.
 - **Governing Equation:** $^{14}_6\text{C} \rightarrow ^{14}_7\text{N} + ^0_{-1}\beta + \bar{\nu}_e$ (where $\bar{\nu}_e$ is an anti-neutrino).
- **Gamma Decay (γ):**
 - **Definition:** Emission of high energy, high frequency electromagnetic radiation from an excited nucleus.
 - **Properties:** No charge; no mass; least ionising radiation.
 - **Penetration:** Highest; significantly reduced by several cm of lead or meters of concrete.
 - **Governing Equation:** $^{60}_{28}\text{Ni-}m \rightarrow ^{60}_{28}\text{Ni} + \gamma$ (where $-m$ indicates a metastable/excited state).
- **Dangers from Nuclear Radiations**
 - **Mechanistic Reasoning:** Ionising radiation interacts with atoms in human cells, causing DNA mutations and increasing cancer risk.
 - **Hazard Levels (Inside vs. Outside Body):**
 - **Alpha:** Highly dangerous inside (high ionisation); skin acts as barrier outside.
 - **Beta:** Moderate ionisation/danger both inside and outside.
 - **Gamma:** Minimal ionisation/danger relative to others, but high penetration allows it to reach internal organs from outside.

[Diagram: Figure B - Setup for investigating radiation penetration using a G-M tube, counter, and different absorber materials like paper, aluminium, and lead] [Diagram: Figure C - Representation of alpha particle emission from a nucleus] [Diagram: Figure D - Representation of beta-minus particle emission (electron) from a nucleus]

9A.2 RATE OF RADIOACTIVE DECAY

- **Probability and Decay**
 - **Definition: Radioactive Decay:** A spontaneous and random process where a nucleus may decay at any moment.
 - **Definition: Decay Constant (λ):** The probability per second that a given nucleus will decay.
 - **Governing Law: Activity (A):** The number of decays per second; proportional to the number of nuclei (N).

- **Equation:** $A = -\frac{dN}{dt} = \lambda N$.
- **Exponential Decay**
 - **Core Theory:** The number of nuclei remaining decreases exponentially over time.
 - **Governing Equation:** $N = N_0 e^{-\lambda t}$.
 - **Activity Equation:** $A = A_0 e^{-\lambda t}$.
 - **Constraints:** N_0 and A_0 are the initial values at $t = 0$.
- **Half-life ($t_{1/2}$)**
 - **Definition:** The time taken for the number of nuclei within a sample (or the activity) to reduce to half its initial value.
 - **Derivation logic:**
 - Set $N = \frac{N_0}{2}$ in $N = N_0 e^{-\lambda t}$.
 - $\frac{1}{2} = e^{-\lambda t_{1/2}}$.
 - $\ln\left(\frac{1}{2}\right) = -\lambda t_{1/2}$.
 - **Final Result:** $t_{1/2} = \frac{\ln 2}{\lambda}$.
 - **Rearranged Form:** $\lambda = \frac{\ln 2}{t_{1/2}}$.
- **Logarithmic Half-life Analysis**
 - **Core Concept:** Linearizing exponential data to find λ .
 - **Governing Equations:**
 - $\ln A = -\lambda t + \ln A_0$.
 - $\ln N = -\lambda t + \ln N_0$.
 - **Graphical Behavior:** A plot of $\ln A$ against t results in a straight line with gradient $-\lambda$ and y-intercept $\ln A_0$.

[Graph: Figure C - Activity (Bq) vs. Time (s) showing an exponential decay curve used to determine half-life] [Graph: Figure D - $\ln A$ vs. Time (s) showing a straight line with a negative gradient equal to λ]

9A.3 FISSION AND FUSION

- **Energy-Mass Equivalence**

- **Governing Law:** Einstein's relationship stating that mass (m) and energy (E) are interchangeable.
- **Equation:** $\Delta E = c^2 \Delta m$.
- **Constants:** $c = 3.00 \times 10^8 \text{ m s}^{-1}$; $1 \text{ u} \approx 1.66 \times 10^{-27} \text{ kg}$.
- **Conversion:** 1 u of mass deficit is equivalent to 931.5 MeV of energy.

- **Mass Deficit (Δm)**

- **Definition:** The difference between the measured mass of a nucleus and the sum of the total masses of its individual constituent nucleons.
- **Mechanistic Reasoning:** Mass is "lost" as it is converted into the energy required to hold the nucleus together.

- **Nuclear Binding Energy (ΔE)**

- **Definition:** The energy used to hold the nucleus together, converted from the mass deficit.
- **Binding Energy per Nucleon:** The total binding energy divided by the number of nucleons (A); indicates nuclear stability.
- **Stability Condition:** Iron-56 ($Fe - 56$) has the highest binding energy per nucleon (approx. 8.8 MeV) and is the most stable isotope.

- **Nuclear Fusion**

- **Definition:** The process where small nuclei combine together to make larger nuclei.
- **Mechanism:** Binding energy per nucleon increases for products (up to $Fe - 56$), releasing energy.
- **Conditions:** Requires extremely high densities and temperatures (e.g., $15 \times 10^6 \text{ K}$ in the Sun) to overcome electrostatic repulsion between nuclei.
- **Example (Proton-Proton Chain):** Four protons eventually fuse into a Helium-4 nucleus.

- **Nuclear Fission**

- **Definition:** The process where larger nuclei break up into smaller nuclei.
- **Mechanism:** Binding energy per nucleon of fission products is higher than the original large nucleus, releasing energy.
- **Induced Fission:** Occurs when a heavy nucleus (e.g., $U - 235$) absorbs a slow-moving "thermal" neutron, becoming unstable ($U - 236$) and splitting.
- **Example:** ${}^{235}_{92}\text{U} + {}^1_0\text{n} \rightarrow {}^{141}_{56}\text{Ba} + {}^{92}_{36}\text{Kr} + 3{}^1_0\text{n} + \text{energy}$.

[Graph: Figure B - Binding energy per nucleon (MeV) vs. Mass number (A) showing fusion region (rising), peak at Fe-56, and fission region (falling)] [Diagram: Figure C - The multi-stage

proton-proton chain fusion reaction] [Diagram: Figure D - Induced nuclear fission of Uranium-235 showing neutron absorption and splitting into daughter nuclei plus neutrons]

9A THINKING BIGGER: MEDICAL TRACERS

- **Application:** Using radioisotopes to monitor body functions.
 - **Technetium-99m** ($Tc - 99m$):
 - **Properties:** Gamma emitter (allows external detection with minimal internal damage); half-life of 6 hours (long enough for medical scans, short enough to limit patient exposure).
 - **Decay:** Produced by the decay of Molybdenum-99 ($t_{1/2} = 66$ hours).
-

CONSOLIDATED EQUATION AND SYMBOL LIST

- **Equations:**
 - $A = -\frac{dN}{dt} = \lambda N$ (Activity and decay constant)
 - $N = N_0 e^{-\lambda t}$ (Nuclei remaining)
 - $A = A_0 e^{-\lambda t}$ (Activity remaining)
 - $t_{1/2} = \frac{\ln 2}{\lambda}$ (Half-life relation)
 - $\ln A = -\lambda t + \ln A_0$ (Linearized decay)
 - $\Delta E = c^2 \Delta m$ (Mass-energy equivalence)
 - $1 \text{ u} = 931.5 \text{ MeV}$ (Energy conversion)
 - **Symbols:**
 - A : Activity (Bq)
 - λ : Decay constant (s^{-1})
 - N : Number of undecayed nuclei
 - $t_{1/2}$: Half-life (s)
 - Δm : Mass deficit/defect (kg or u)
 - ΔE : Binding energy (J or MeV)
 - c : Speed of light ($3.00 \times 10^8 \text{ m s}^{-1}$)
 - α, β, γ : Types of radiation
-

ULTRA-CONDENSED FINAL SUMMARY

- **Radiation:** Alpha ($+2e$, high ionisation, low penetration), Beta ($-1e$, mod ionisation/penetration), Gamma (0 , low ionisation, high penetration).
- **Decay Law:** Random/spontaneous; follows exponential decay $N = N_0 e^{-\lambda t}$. Half-life $t_{1/2} = \ln 2 / \lambda$.
- **Stability:** Measured by binding energy per nucleon; peaks at Iron-56.
- **Fusion:** Small nuclei combine; occurs in stars; requires high temperature/density.
- **Fission:** Heavy nuclei split; induced by neutron absorption; releases energy as products are more stable.
- **Equivalence:** Mass and energy are related by $E = mc^2$; mass deficit in nuclei equals binding energy.

TOPIC 10: OSCILLATIONS

10A OSCILLATIONS

10A.1 SIMPLE HARMONIC MOTION (SHM)

- **Definition / Core Concept**
 - **Oscillate:** To undertake continuously repeated movements.
 - **Period (T):** The time taken for one complete oscillation.
 - **Frequency (f):** The number of oscillations per unit time; $f = \frac{1}{T}$.
 - **Amplitude (A):** The maximum displacement from the equilibrium position.
- **Governing Law: Restoring Force**
 - The force acting on the object is a restoring force (F), which acts to return the object to its center position.
 - **Equation:** $F = -kx$.
 - **Explanation:** The force is proportional to the displacement (x) but acts in the opposite direction.
- **Conditions for SHM**
 - An oscillation is simple harmonic if the acceleration (a) is proportional to the displacement (x) and directed toward the equilibrium position.
 - **Mechanism:** From Newton's second law ($F = ma$), because $F \propto -x$, then $a \propto -x$.
- **Assumptions and Constraints**
 - **Isochronous:** The period (T) of SHM is independent of the amplitude.
 - **Small Angle Approximation:** For a pendulum, SHM only applies for small swings where the restoring force is proportional to horizontal displacement.
- **Special Cases: Pendulum and Mass-Spring System**

- **Mass-Spring Period:** $T = 2\pi\sqrt{\frac{m}{k}}$.
 - m = mass (kg), k = spring constant (Nm^{-1}).
- **Pendulum Period:** $T = 2\pi\sqrt{\frac{l}{g}}$.
 - l = length (m), g = acceleration due to gravity ($9.81ms^{-2}$).
 - Significance: Period depends only on length and gravity, not mass or amplitude.

[Diagram: Figure B - Restoring force in a mass-spring system at equilibrium ($F = 0$), stretched ($F = -kx$), and double-stretched ($F = -k(2x)$) positions]

10A.2 SHM MATHEMATICS

- **Angular Velocity (ω) and SHM**
 - SHM is the projection of uniform circular motion onto a diameter.
 - **Equation:** $\omega = \frac{\theta}{t} = \frac{2\pi}{T} = 2\pi f$.
 - Units: Radians per second ($rads^{-1}$).
- **Displacement Equation**
 - If the object starts at maximum displacement (A) at $t = 0$:
 - **Equation:** $x = A \cos(\omega t)$.
- **Velocity and Acceleration Derivations**
 - Velocity (v) is the rate of change of displacement ($\frac{dx}{dt}$):
 - **Equation:** $v = -A\omega \sin(\omega t)$.
 - Maximum velocity ($v_{max} = A\omega$) occurs at equilibrium ($x = 0$).
 - Acceleration (a) is the rate of change of velocity ($\frac{dv}{dt}$):
 - **Equation:** $a = -A\omega^2 \cos(\omega t)$.
 - Maximum acceleration ($a_{max} = A\omega^2$) occurs at maximum displacement ($x = A$).
- **Defining Equation of SHM**
 - Substituting $x = A \cos(\omega t)$ into the acceleration equation:
 - **Equation:** $a = -\omega^2 x$.
 - **Cause-Effect:** Acceleration is always proportional to displacement and in the opposite direction.
- **SHM Constant (k)**

- From $a = -\frac{k}{m}x$, it follows that $\omega^2 = \frac{k}{m}$.

[Graph: Figure C - Displacement, Velocity, and Acceleration vs Time. Displacement is a cosine curve, Velocity is a negative sine curve, Acceleration is a negative cosine curve]

10A.3 SHM ENERGY

- **Core Concept: Energy Conservation**
 - In a closed (undamped) system, total energy (E_{total}) remains constant.
 - **Mechanism:** Energy is continuously transferred between kinetic energy (E_k) and potential energy (E_p).
- **Energy States**
 - **Equilibrium ($x = 0$):** Potential energy is zero; Kinetic energy is at maximum ($E_k = \frac{1}{2}mv_{max}^2$).
 - **Amplitude ($x = \pm A$):** Kinetic energy is zero (instantaneously stationary); Potential energy is at maximum.
- **Energy Transfers**
 - Potential energy can be Gravitational (mgh) for a pendulum or Elastic ($\frac{1}{2}kx^2$) for a spring.
 - The sum $E_k + E_p$ is constant at every point in the oscillation.

[Graph: Figure A - Energy vs Displacement. Total energy is a horizontal line; E_k is an inverted parabola peaking at $x = 0$; E_p is a parabola peaking at $x = \pm A$]

10A.4 RESONANCE AND DAMPING

- **Free and Forced Oscillations**
 - **Natural Frequency (f_0):** The frequency at which a system oscillates when left to vibrate freely.
 - **Driving Frequency (f):** The frequency of an external force applied to a system.
 - **Forced Oscillation:** When an external driving force is applied, the system is forced to vibrate at the driving frequency.
- **Resonance**
 - **Definition:** The phenomenon where the driving frequency matches the natural frequency ($f = f_0$), resulting in maximum energy transfer and a very large amplitude of vibration.

- **Significant Results:** Can cause structural damage (e.g., breaking glass, bridge collapse).
- **Damping**
 - **Definition:** The loss of energy in each oscillation due to resistive forces (like friction or air resistance), reducing the amplitude over time.
 - **Light Damping (Underdamping):** Amplitude decreases exponentially over many cycles; the period remains constant.
 - **Critical Damping:** The system returns to the equilibrium position in the quickest possible time without overshooting.
 - **Overdamping:** The system returns very slowly to equilibrium without oscillating (e.g., thick treacle).
- **Effect of Damping on Resonance**
 - Increasing damping reduces the maximum amplitude at resonance.
 - Heavier damping shifts the resonant peak to a slightly lower frequency.

[Graph: Figure A - Amplitude vs Frequency showing a sharp peak at resonance (f_0)] [Graph: Figure D - Displacement vs Time for different damping: Underdamped (oscillatory decay), Critically damped (fastest non-oscillatory return), Overdamped (slow return)]

10A THINKING BIGGER: TUNED MASS DAMPER (TAIPEI 101)

- **Problem:** High-rise buildings sway due to wind or earthquakes.
 - **Solution: Tuned Mass Damper (TMD)**
 - **Mechanism:** A massive pendulum (728 tonnes in Taipei 101) positioned near the top of the building.
 - **Physics:** The TMD is "tuned" to the building's natural frequency but oscillates out of phase with the building's movement, absorbing energy and reducing sway by up to 40%.
 - **Damping:** Hydraulic pistons (viscous dampers) restricted movement and dissipated the absorbed energy as heat.
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CONSOLIDATED EQUATION AND SYMBOL LIST

- **Basic SHM**
 - $f = \frac{1}{T}$ (Frequency-Period relation)
 - $F = -kx$ (Restoring force)
 - $a = -\omega^2 x$ (Defining equation of SHM)
- **Angular Velocity and Time**

- $\omega = \frac{2\pi}{T} = 2\pi f$ (Angular velocity)
 - $T = 2\pi\sqrt{\frac{m}{k}}$ (Mass-spring period)
 - $T = 2\pi\sqrt{\frac{l}{g}}$ (Pendulum period)
 - **Motion Functions**
 - $x = A \cos(\omega t)$ (Displacement)
 - $v = -A\omega \sin(\omega t)$ (Velocity)
 - $a = -A\omega^2 \cos(\omega t)$ (Acceleration)
 - $v_{max} = A\omega$ (Maximum velocity)
 - $a_{max} = A\omega^2$ (Maximum acceleration)
 - **Resonance**
 - $f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$ (Natural frequency of mass-spring)
 - **Symbols**
 - T : Period (s)
 - f : Frequency (Hz)
 - A : Amplitude (m)
 - x : Displacement (m)
 - ω : Angular velocity ($rad s^{-1}$)
 - k : Stiffness / Spring constant ($N m^{-1}$)
 - m : Mass (kg)
 - l : Length of pendulum (m)
 - g : Acceleration due to gravity ($9.81 m s^{-2}$)
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ULTRA-CONDENSED FINAL SUMMARY

- **SHM Condition:** Acceleration is proportional to negative displacement ($a \propto -x$).
- **Restoring Force:** Always directed toward equilibrium ($F = -kx$).
- **Isochronicity:** Period (T) is independent of Amplitude (A).
- **Energy:** Constant total energy toggles between E_k (max at center) and E_p (max at edges).
- **Resonance:** Occurs when driving frequency matches natural frequency; maximizes amplitude.
- **Damping:** Removes energy. Underdamping (decaying oscillations), Critical (fastest return), Overdamping (slow crawl).

- **Structural App:** Tuned Mass Dampers use resonance/damping principles to stabilize skyscrapers against wind/seismic loads.

TOPIC 11: ASTROPHYSICS AND COSMOLOGY

11A GRAVITATIONAL FIELDS

11A.1 GRAVITATIONAL FORCES

- **Definition / Core Concept**
 - **Gravity:** A universal force of attraction where every particle with mass attracts every other particle with mass.
 - **Accrete:** To grow slowly by attracting and joining with many small pieces of rock and dust due to the force of gravity.
- **Governing Law: Newton's Law of Universal Gravitation**
 - The gravitational force (F) between two masses (m_1, m_2) is directly proportional to the product of their masses and inversely proportional to the square of the distance (r) between their centers.
 - **Equation:**
$$F = \frac{Gm_1m_2}{r^2}.$$
 - **Gravitational Constant (G):** $6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}.$
- **Conceptual and Mechanistic Explanation**
 - Gravity drives the formation of stars from gas clouds (e.g., Carina nebula).
 - As particles gather into larger bodies, the gravitational pull increases, leading to the formation of planets or stars.
 - Weight ($W = mg$) is the manifestation of the gravitational force exerted by a large body on a mass (m).
- **Assumptions and Limitations**
 - Distance (r) must be measured between the centers of gravity of the two masses.
 - Acceleration is inversely proportional to the square of the distance, so standard equations of motion for uniform acceleration (a) do not apply over long distances.
- **Equations and Derivations**
 - **Field Strength from Force:** $g = \frac{F}{m}.$
 - **Orbital Velocity (v):** For a circular orbit, gravity provides the centripetal force:
$$\frac{GMm}{r^2} = \frac{mv^2}{r}.$$

- **Orbital Period (T):** Since $v = \frac{2\pi r}{T}$, then $T = \sqrt{\frac{4\pi^2 r^3}{GM}}$.
- **Mass of Central Body (M):** Derived from orbit: $M = \frac{rv^2}{G}$.
- **Practical Significance: Weighing the Earth**
 - The mass of the Earth ($M_E \approx 6 \times 10^{24}$ kg) can be determined using the Moon's orbital radius and period.

[Diagram: Figure D - Weighing the Earth by observing the Moon's orbital parameters]

11A.2 GRAVITATIONAL FIELDS

- **Definition / Core Concept**
 - **Gravitational Field:** A region of spacetime which is curved, causing particles to experience an accelerating force.
 - **Gravitational Potential (V_{grav}):** The amount of work done per unit mass to move an object from an infinite distance to that point in the field.
- **Governing Principles: Radial Fields**
 - Any mass generates a radial gravitational field that is always attractive.
 - Field lines show the direction of force and point towards the center of the mass.
- **Equations and Derivations**
 - **Gravitational Field Strength (g):** $g = \frac{GM}{r^2}$.
 - **Gravitational Potential (V_{grav}):** $V_{grav} = -\frac{GM}{r}$.
 - **Potential Gradient:** $g = -\frac{dV_{grav}}{dx}$.
 - **Potential Energy Change:** $\Delta E_p = m\Delta V_{grav}$.
- **Mechanistic Reasoning: Negative Potential**
 - Potential is zero at infinity ($r = \infty$).
 - Because gravity is attractive, work must be done to move a mass away from a field; therefore, potential increases as you move away and is always negative near a mass.
- **Special Case: Field Comparison**
 - **Similarity:** Both gravitational and electric fields follow inverse square laws for force.
 - **Difference:** Gravitational forces are only attractive; electric forces can be attractive or repulsive.
 - **Shielding:** Electric fields can be shielded; gravitational fields cannot.

[Diagram: Figure A - Radial gravitational field around a point mass showing field lines pointing inwards]

11A THINKING BIGGER: MONITORING VOLCANOES

- **Application:** Using gravity and electromagnetic monitoring to predict eruptions.
 - **Mechanism:**
 - **Microgravity:** Measures minute changes in the Earth's gravity to detect movements of magma and gases within a volcano.
 - **Vesiculation:** The formation of gas bubbles in a magma chamber, which changes the density of the surroundings and is detectable via microgravity.
 - **Electrical Resistivity:** Changes in magma movement and type alter the local electromagnetic fields.
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11B SPACE

11B.1 STARSHINE

- **Definition / Core Concept**
 - **Black Body Radiator:** A theoretical body which completely absorbs all radiation that lands on it and reflects none.
 - **Luminosity (L):** The rate at which energy of all types is radiated by an object in all directions (output power).
- **Governing Law: Stefan-Boltzmann Law**
 - Luminosity is proportional to the surface area (A) and the fourth power of absolute temperature (T).
 - **Equation:** $L = \sigma AT^4$.
 - **For a Sphere:** $L = 4\pi r^2 \sigma T^4$.
 - **Stefan-Boltzmann Constant (σ):** $5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$.
- **Governing Law: Wien's Law**
 - The peak wavelength (λ_{max}) of radiation from a black body is inversely proportional to its absolute temperature.
 - **Equation:** $\lambda_{max} T = 2.898 \times 10^{-3} \text{ m K}$.
- **Conceptual Explanation: Radiation Curves**
 - As temperature increases, the peak wavelength shifts towards the shorter (bluer) end of the spectrum.
 - Hotter stars emit more energy at all wavelengths than cooler stars.

[Graph: Figure D - Black body radiation curves showing Energy Output vs Wavelength for different temperatures]

11B.2 STELLAR CLASSIFICATION

- **Spectral Classes**
 - Stars are grouped by their effective temperature.
 - **Classification (Hottest to Coolest):** O, B, A, F, G, K, M.
 - **Mnemonic:** "Only Brilliant Astronomers Fix Grave Kepler Mistakes".
 - **Class O:** Blue ($> 28,000$ K); **Class M:** Red ($2,000$ – $3,500$ K).
- **Hertzsprung-Russell (H-R) Diagram**
 - A plot of luminosity (L) against absolute temperature (T).
 - **Main Sequence:** A diagonal line from top-left to bottom-right where stars spend the majority of their lifetime fusing hydrogen.
 - **Red Giants/Supergiants:** Located at the top right (cool but very large and luminous).
 - **White Dwarfs:** Located at the bottom left (hot but very small and dim).
- **Stellar Evolution Pathways**
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- **Significant Results**
 - **Supernova:** The explosion of a large mass star at the end of its lifetime when it becomes unstable.
 - **Black Hole:** A region of space-time in which gravity is so strong that nothing, including EM radiation, can escape.
 - **Neutron Star:** A small, very dense star composed of neutrons; one of the possible conclusions to the life of a large mass star.

[Graph: Figure C - Hertzsprung-Russell diagram showing Luminosity vs Temperature with identified regions for Main Sequence, Giants, and White Dwarfs]

11B.3 DISTANCES TO THE STARS

- **Distance Units**
 - **Astronomical Unit (AU):** Radius of Earth's orbit around the Sun ($1 AU \approx 1.5 \times 10^{11}$ m).
 - **Light Year (ly):** Distance light travels in one year ($\approx 9.46 \times 10^{15}$ m).
 - **Parsec (pc):** The distance to a star that has a parallax angle of 1 arcsecond ($1/3600$ degrees).
 - **Conversion:** $1 pc \approx 3.09 \times 10^{16}$ m ≈ 3.26 light years.
- **Trigonometric Parallax**

- **Core Concept:** Measuring the apparent shift of a nearby star against a background of distant stars over a six-month period.
- **Governing Equation:** $d = \frac{r}{\tan \theta} \approx \frac{r}{\theta}$.
- **Simplified Equation:** $d = \frac{1}{\theta}$ (where d is in parsecs and θ is in arcseconds).
- **Intensity and Standard Candles**
 - **Inverse Square Law for Radiation:** $I = \frac{L}{4\pi d^2}$.
 - **Standard Candle:** An astronomical object (like a Cepheid variable) of known luminosity used to determine distances.

[Diagram: Figure C - Right-angled triangle formed by Earth-Sun distance (r), distance to star (d), and parallax angle (θ)]

CONSOLIDATED EQUATION AND SYMBOL LIST

- **Gravitation**
 - $F = \frac{Gm_1m_2}{r^2}$ (Universal Gravitation)
 - $g = \frac{F}{m} = \frac{GM}{r^2}$ (Gravitational Field Strength)
 - $V_{grav} = -\frac{GM}{r}$ (Gravitational Potential)
 - $g = -\frac{dV_{grav}}{dx}$ (Potential Gradient)
 - $\Delta E_p = m\Delta V_{grav}$ (Potential Energy Change)
 - $T = \sqrt{\frac{4\pi^2 r^3}{GM}}$ (Orbital Period)
- **Stars and Distances**
 - $L = \sigma AT^4$ (Stefan-Boltzmann Law)
 - $\lambda_{max}T = 2.898 \times 10^{-3} \text{ m K}$ (Wien's Law)
 - $d \text{ (pc)} = \frac{1}{\theta \text{ (arcsec)}}$ (Distance via Parallax)
 - $I = \frac{L}{4\pi d^2}$ (Inverse Square Law for Intensity)
- **Symbols and Constants**

- G : Gravitational constant ($6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$)
 - σ : Stefan-Boltzmann constant ($5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$)
 - L : Luminosity (W)
 - V_{grav} : Gravitational potential ($J \text{ kg}^{-1}$)
 - λ_{max} : Peak wavelength (m)
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ULTRA-CONDENSED FINAL SUMMARY

- **Gravity:** Always attractive inverse-square force ($F \propto 1/r^2$). Potential (V) is energy per unit mass, zero at infinity, and always negative.
- **Orbits:** Centripetal force is provided by gravity; allows "weighing" of stars and planets.
- **Star Properties:** Luminosity depends on T^4 and Area (Stefan-Boltzmann). Peak wavelength relates to $1/T$ (Wien).
- **Stellar Evolution:** Stars follow paths on the H-R diagram based on mass. Low mass ends as White Dwarfs; high mass ends as Neutron Stars or Black Holes.
- **Distances:** Trigonometric parallax ($d = 1/\theta$) works for nearby stars; standard candles are used for greater distances.